Homework Solution
Chapter 4: Nominal and Effective Interest Rates

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Exercise 1: Nominal Interest Rate on a Student Savings Account

Solution:

Using $r = (\text{rate per period}) \times (\text{number of periods})$:

Part 1. With 12 months per year:

$$\begin{aligned} r &= 0.75\% \times 12 \\ &= 9.00\% \text{ per year} \end{aligned}$$

Part 2. With 3 months per quarter:

$$\begin{aligned} r &= 0.75\% \times 3 \\ &= 2.25\% \text{ per quarter} \end{aligned}$$

Exercise 2: Nominal to Effective Annual Rate on a Credit Card

Solution:

Part 1. $r = 15\%$ per year, $m = 12$ (monthly compounding).

Part 2. Using

$$i_a = \left(1 + \frac{r}{m}\right)^m - 1$$

$$\begin{aligned} i_a &= \left(1 + \frac{0.15}{12}\right)^{12} - 1 \\ &= (1.0125)^{12} - 1 \\ &\approx 16.08\% \end{aligned}$$

Part 3. Since the competing card's effective annual rate is 15.5%, and $16.08\% > 15.5\%$, **the original card (**15% **nominal, compounded monthly) is more expensive** for the cardholder.

Exercise 3: Comparing Effective Rates Across Compounding Periods

Solution:

Part 1. Effective rate per period $= r/m$:

- CD 1 (semi-annual, $m = 2$): $8\%/2 = 4.00\%$ per period
- CD 2 (quarterly, $m = 4$): $8\%/4 = 2.00\%$ per period
- CD 3 (monthly, $m = 12$): $8\%/12 \approx 0.6667\%$ per period

Part 2. Using $i_a = (1 + r/m)^m - 1$:

- CD 1: $(1.04)^2 - 1 = 8.16\%$
- CD 2: $(1.02)^4 - 1 \approx 8.24\%$
- CD 3: $(1 + 0.08/12)^{12} - 1 \approx 8.30\%$

Part 3. For a fixed nominal rate, the effective annual rate **increases as the compounding frequency increases**, since interest is added to the balance more often and begins earning interest on itself sooner. This trend approaches, but never exceeds, the continuous-compounding limit $e^r - 1$.

Exercise 4: Comparing a Bank's Compounded Rate to a Competitor's Stated Rate

Solution:

Part 1. First bank: $r = 11\%$, $m = 12$:

$$i_a = \left(1 + \frac{0.11}{12}\right)^{12} - 1 \approx 11.57\%$$

Part 2. Second bank: interest is applied only once per year, so the stated 11.4% is already an effective annual rate; $i_a = 11.40\%$.

Part 3. Since $11.40\% < 11.57\%$, **the supplier should choose the second bank**, as it results in a lower true annual interest cost.

Exercise 5: Continuous Compounding on a Long-Term Reserve Fund

Solution:

Part 1. Under continuous compounding, $i = e^r - 1$:

$$i = e^{0.06} - 1 \approx 6.18\%$$

Part 2. $F = Pe^{rt}$:

$$\begin{aligned} F &= 50{,}000\, e^{0.06(8)} \\ &= 50{,}000\, e^{0.48} \\ &\approx \$80{,}803.72 \end{aligned}$$

Part 3. $r = 6.15\%$, $m = 12$, $t = 8$ years, $k = 96$ months:

$$F = 50{,}000 \left(1 + \frac{0.0615}{12}\right)^{96} \approx \$81{,}676.52$$

Since \$81,676.52 $>$ \$80,803.72, **the competing fund (monthly compounding at** 6.15%**) leaves the hospital with more money** at the end of 8 years.

Exercise 6: Evaluating Two Equipment Financing Offers

Solution:

Part 1. Bank A: $r = 8.4\%$, $m = 12$:

$$i_a = \left(1 + \frac{0.084}{12}\right)^{12} - 1 \approx 8.73\%$$

Part 2. In-house financing: 2.1% per quarter, $m = 4$:

$$i_a = (1.021)^4 - 1 \approx 8.67\%$$

Part 3. Bank B's rate is already effective: $i_a = 8.70\%$. Ranking cheapest to most expensive:

$$\text{In-house } (8.67\%) \;<\; \text{Bank B } (8.70\%) \;<\; \text{Bank A } (8.73\%)$$

The in-house financing arm offers the lowest true annual cost, followed closely by Bank B, with Bank A being the most expensive.

Exercise 7: Present Worth of a Future Car Purchase

Solution:

Part 1. $r = 10\%$, $m = 12$, and $k = 3 \times 12 = 36$ months.

Part 2. Using $P = F(1 + r/m)^{-k}$:

$$P = 60{,}000\left(1 + \frac{0.10}{12}\right)^{-36} \approx \$44{,}504.38$$

Part 3. Since Ali's available \$45,000 exceeds the required \$44,504.38, **his \$45,000 is sufficient**, leaving a surplus of

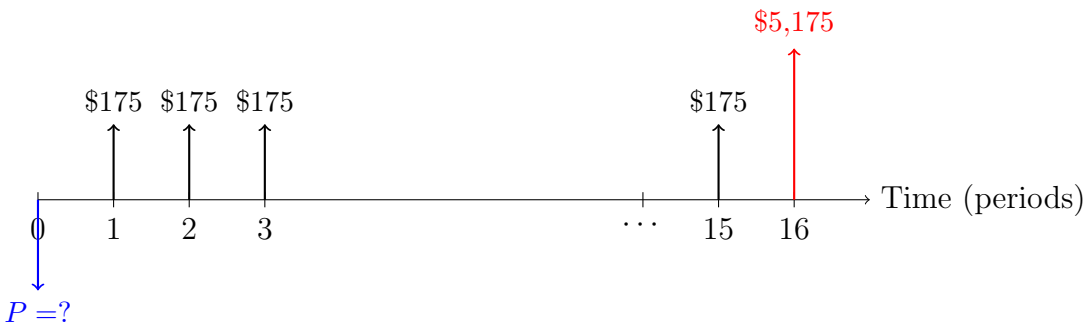
$$\$45{,}000 - \$44{,}504.38 = \$495.62$$

Exercise 8: Pricing a Corporate Bond Given a Required Yield

Solution:

Part 1. $F = \$5{,}000$, annual coupon $= 7\% \times 5{,}000 = \$350$, so $C = \$350$ per year (semi-annual coupon payment of \$175); $\lambda = 9\%$, $m = 2$, $n = 8 \times 2 = 16$ semi-annual periods.

Part 2. The cashflow diagram is shown below.



Part 3. Using

$$P = F\left(1 + \frac{\lambda}{m}\right)^{-n} + \frac{C}{m}\left[\frac{1 - (1 + \lambda/m)^{-n}}{\lambda/m}\right]$$

with $\lambda/m = 0.045$:

$$P = 5{,}000(1.045)^{-16} + 175\left[\frac{1 - (1.045)^{-16}}{0.045}\right] \approx \$4{,}438.30$$

Part 4. Since the coupon rate (7%) is lower than the yield to maturity (9%), the bond **trades at a discount**, which is consistent with the computed price (\$4,438.30) being below the \$5,000 face value.

Exercise 9: Premium, Discount, and Par Pricing Across Two Bonds

Solution:

Part 1. Bond A's coupon rate (6%) is below the yield (7.5%), so Bond A will trade at

a **discount**. Bond B's coupon rate (9%) is above the yield (7.5%), so Bond B will trade at a **premium**.

Part 2. $F = 1{,}000$, $\lambda/m = 0.0375$, $n = 20$, semi-annual coupon $= 30$:

$$P_A = 1{,}000(1.0375)^{-20} + 30\left[\frac{1-(1.0375)^{-20}}{0.0375}\right] \approx \$895.78$$

Part 3. Semi-annual coupon $= 45$:

$$P_B = 1{,}000(1.0375)^{-20} + 45\left[\frac{1-(1.0375)^{-20}}{0.0375}\right] \approx \$1{,}104.22$$

Part 4. As the required yield rises to 10%, **both bond prices will decrease**. A higher required yield means future cash flows (coupons and face value) are discounted more heavily, which lowers the present value regardless of the bond's own coupon rate; the effect applies to Bond A and Bond B alike, though the size of the price change may differ between them.

Exercise 10: Investment Growth Under a Time-Varying, Multi-Structure Rate Agreement

Solution:

Part 1. Year 1: $r = 8\%$, $m = 4$:

$$i_1 = (1.02)^4 - 1 \approx 8.24\%$$

Part 2. Year 2: $r = 10\%$, $m = 12$:

$$i_2 = \left(1 + \frac{0.10}{12}\right)^{12} - 1 \approx 10.47\%$$

Part 3. Year 3 rate is already effective: $i_3 = 9\%$. Chaining all three years:

$$\begin{aligned} F &= 15{,}000\,(1+i_1)(1+i_2)(1+i_3) \\ &= 15{,}000(1.0824)(1.1047)(1.09) \\ &\approx \$19{,}550.95 \end{aligned}$$

Part 4. The equivalent constant effective annual rate i_{eq} must satisfy:

$$15{,}000\,(1+i_{\text{eq}})^3 = 19{,}550.95$$